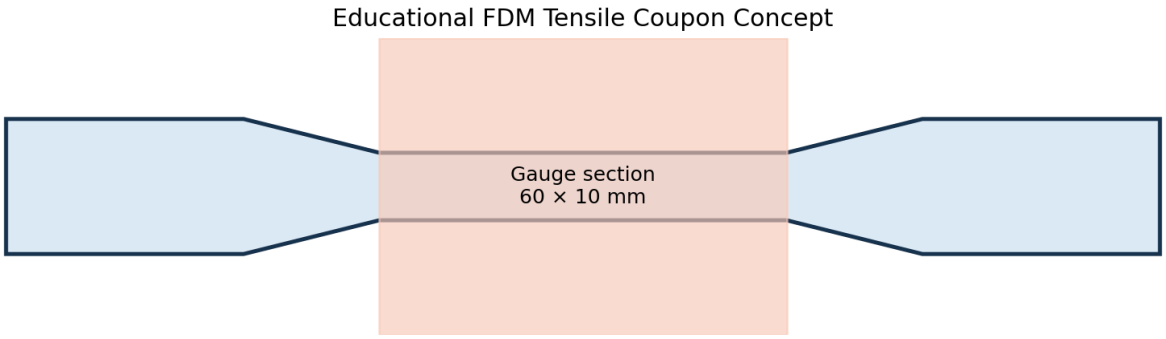
# FDM Tensile Coupon Design and DOE Optimization

## Coursework-based academic project applying MCE 515 Additive Manufacturing concepts




Conclusion. The simulated full-factorial DOE identified 0.16 mm layer height, 35 mm/s print speed, and 100% infill as the strongest evaluated condition. Its mean response was 46.0 MPa versus 40.0 MPa for the baseline, a 15.0% simulated improvement.

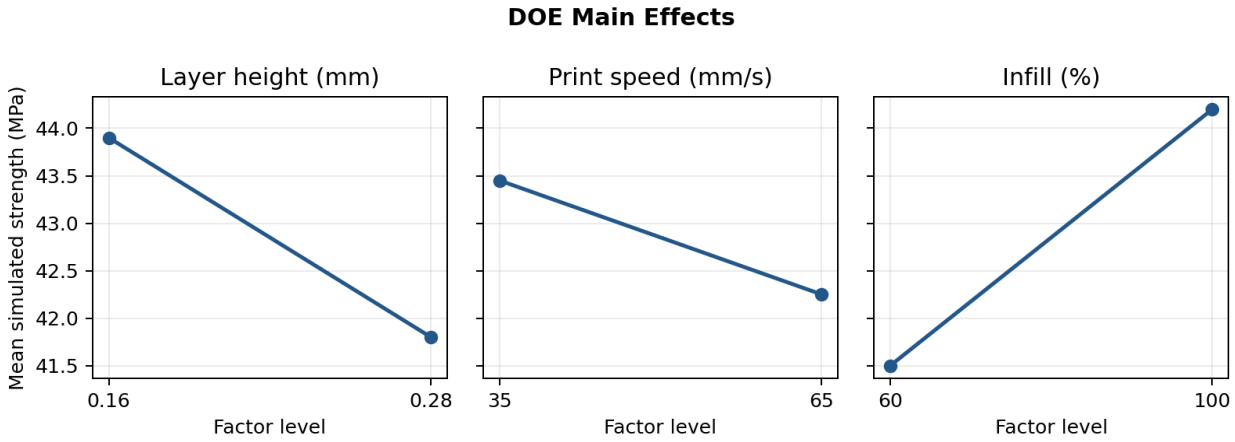
# 1 Project scope

This academic portfolio project applies concepts from MCE 515 Additive Manufacturing at Cleveland State University. The coupon geometry and simulated dataset were developed after completing the course. No coupons were printed or tensile-tested, and the response values are not physical test results.

# 2 Design and DOE method

The educational dog-bone-style coupon has a 170 mm overall length, 60 mm gauge length, 10 mm gauge width, and 4 mm thickness. The $2^3$ full-factorial design evaluates layer height, print speed, and infill density at two levels, with three simulated replicates for each of eight conditions.

| Layer | Speed | Infill | Mean MPa | Stdev |
|---|---|---|---|---|
| 0.16 | 35 | 60% | 43.0 | 0.30 |
| 0.16 | 35 | 100% | 46.0 | 0.30 |
| 0.16 | 65 | 60% | 41.8 | 0.30 |
| 0.16 | 65 | 100% | 44.8 | 0.30 |
| 0.28 | 35 | 60% | 41.2 | 0.30 |
| 0.28 | 35 | 100% | 43.6 | 0.30 |
| 0.28 | 65 | 60% | 40.0 | 0.30 |
| 0.28 | 65 | 100% | 42.4 | 0.30 |




# 3 Engineering interpretation

Within this simulated study, lower layer height, lower print speed, and higher infill increase predicted tensile strength. The results are useful for demonstrating DOE structure and parameter screening, but they cannot establish material properties. Physical replication, randomized run order, environmental control, and measured uncertainty are required before making manufacturing decisions.